CSE 3200: System Development Project

Horizons: A Gameplay-Based Autism Spectrum Disorder Screening System

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Acknowledgment

All praise to the Almighty Allah, whose mercy and guidance made it possible to bring this project to completion. We express sincere gratitude to our supervisor, Dr. Md. Aminul Haque Akhand, for his patient direction, critical feedback and steady encouragement from the earliest design discussions through to the final submission.

We are grateful to the faculty of the Department of Computer Science and Engineering, Khulna University of Engineering & Technology, for the academic environment that shaped this work. Our classmates contributed through many informal discussions that helped clarify difficult design decisions and the families and children who participated in the pilot validation study gave their time generously; without their cooperation, the clinical evaluation would not have been possible.

We also acknowledge the open-source communities behind Next.js, Bun, Tailwind CSS, ShadCN UI, Framer Motion, Tone.js, the Neon serverless PostgreSQL platform and Better Auth. Without this freely available tooling, reaching production quality within an academic timeline would not have been achievable. Finally, we thank the clinical researchers whose published work on ADOS-2, discrete trial training game design and the EmoGalaxy emotion screening study provided the scientific foundation that the Horizons scoring model is built on.

Authors

Abstract

Autism Spectrum Disorder (ASD) is a neurodevelopmental condition marked by persistent differences in social communication and by restricted, repetitive patterns of behaviour. Validated direct-observation instruments such as ADOS-2 remain inaccessible to many families because they require trained clinicians, controlled environments and considerable administration time. This project presents *Horizons*, a browser-based serious game that collects structured behavioural signals from children aged 3 to 10 across six interconnected gameplay chapters. The chapters address social communication, restricted and repetitive behaviours, sensory processing and pretend play, drawing from the domain structure of ADOS-2, DSM-5 and M-CHAT-R/F. For every task the game records response time, selection accuracy, attempt count and pointer movement trajectory, which feed a penalty-based scoring engine that computes weighted domain scores and a four-level combined risk classification.

The system runs on Next.js 15 with the Bun runtime, PostgreSQL hosted on the Neon serverless platform via `postgres.js`, Better Auth for role-based researcher authentication, Tone.js for real-time Web Audio sound synthesis and Tailwind CSS v4 with ShadCN UI for the interface. No image or pre-recorded audio files are shipped; visual content is rendered entirely from emoji and CSS and every game sound is synthesised at runtime by Tone.js. A public-facing platform provides landing, about, contact and survey pages. At session completion the scoring engine generates an HMAC-SHA256 signed caregiver report link that can be forwarded directly to a healthcare provider. An authenticated researcher dashboard, secured by Better Auth, delivers per-session domain charts, pointer heatmaps, red-flag summaries and JSON or CSV data exports. An automated test suite of 312 tests covers scoring logic, database queries, API routes and the full scoring pipeline.

Pilot validation with 24 children, including 8 with confirmed ASD diagnoses and 16 neurotypical controls, produced a sensitivity of 87.5% and specificity of 81.3% at the medium risk threshold, figures consistent with prior single-domain emotion screening studies. Horizons is not a diagnostic instrument; it is a screening aid designed to support, not replace, evaluation by qualified healthcare professionals.

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CHAPTER I

Introduction

1.1 Introduction

Autism Spectrum Disorder (ASD) affects roughly 1 in 36 children in the United States [1] and between 1 and 2 per cent of children globally [2]. Presentations vary widely across social communication profiles and show differing degrees of restricted or repetitive behaviour, which means that a single screening pathway cannot suit all families or clinical contexts. Validated direct-observation instruments, particularly ADOS-2 [3], deliver reliable behavioural data but demand trained clinicians, structured assessment environments and significant administration time. Those demands put them beyond reach for many families in resource-limited settings.

Serious games offer a different angle on this problem. A well-designed game can embed validated behavioural tasks within familiar play interactions, capturing structured observational data without the formality of a clinical appointment. Irani et al. [4] showed that a mobile emotion recognition game reached 93% screening accuracy using support vector machine classification, confirming that gameplay response data can carry clinically meaningful signal.

Horizons was built around that premise. The system is a six-chapter browser-based serious game that collects multi-domain behavioural data from children aged 3 to 10, running on any modern-browser device and requiring no installation, dedicated hardware or clinician presence beyond basic supervision.

1.2 Background and Problem Statement

Early ASD screening tools differ substantially in validity, accessibility and ecological validity. Caregiver questionnaires such as M-CHAT-R/F [5] are practical and widely used, but they depend on accurate parental recall and are therefore vulnerable to observer bias. Direct-observation instruments such as ADOS-2 are more accurate but expensive and time-consuming to administer. Tablet and browser-based serious games have emerged as a middle ground; however, most existing implementations run on proprietary platforms, require application installation, cover only a narrow age range or do not collect the timestamped multi-domain behavioural data that rigorous screening requires [6].

The core problem this project addresses is as follows: how can a production-standard, browser-delivered serious game simultaneously provide an engaging experience for children aged 3 to 10 and collect clinically meaningful, multi-domain ASD screening signals without requiring specialised equipment or professional administration?

1.3 Objectives

The primary objectives of this project are:

1. To design and implement a browser-based serious game spanning six chapters that together cover the four ASD indicator domains defined by ADOS-2, DSM-5 and M-CHAT-R/F.

2. To develop a penalty-based multi-domain scoring engine grounded in three peer-reviewed research frameworks: ADOS-2 domain weighting, DTT-based game scoring and the EmoGalaxy task accuracy formula.

3. To implement automatic red-flag detection with multiplicative severity weighting for clinically significant behavioural patterns.

4. To build a researcher dashboard with Better Auth role-based authentication, per-session analytics, pointer heatmaps and JSON or CSV exports.

5. To generate HMAC-SHA256 signed caregiver report links at session completion that a healthcare provider can access without researcher involvement.

6. To validate the combined risk classifier against a pilot cohort that includes children with confirmed ASD diagnoses.

1.4 Scope

Horizons encompasses the full child-facing gameplay flow across six chapters, the scoring and risk-classification engine, public platform pages, the authenticated researcher dashboard and the HMAC-signed caregiver report generation. The principal technologies employed are:

- **Next.js 15 (App Router):** Server components, file-based routing and API route handlers for clean separation of client and server logic.
- **Bun runtime:** Built-in test runner, native package resolution and the `bun` package manager.
- **PostgreSQL via Neon:** Serverless PostgreSQL with the `postgres.js` driver; schema migration runs automatically on every server start.
- **Better Auth:** Open-source email and password authentication with researcher and admin roles, HttpOnly session cookies and in-memory rate limiting.
- **Tailwind CSS v4 with `@theme` tokens:** Design-token-driven theming without raw CSS files beyond `globals.css`.
- **ShadCN UI:** Radix-based accessible component library.
- **Framer Motion:** Physics-based animation with sensory-intensity gating.
- **Tone.js:** Real-time Web Audio synthesis; all game sounds including sensory test stimuli are generated at runtime with zero audio files shipped.
- **@dnd-kit:** Pointer and touch drag-and-drop for sorting tasks.

- **Recharts:** Dashboard charting library.
- **Zustand v5:** Persisted client-side session state.
- **Resend:** Transactional email delivery for contact form notifications.
- **Vercel Analytics:** Zero-configuration visitor analytics.

Out of scope are clinical diagnosis, longitudinal multi-session tracking, medical-device certification and durable personal identity records.

1.5 Unfamiliarity of the Problem/Topic/Solution

Several serious game systems for ASD appear in the research literature, yet none currently combines all of the following characteristics in a single publicly available implementation:

- Browser delivery requiring no installation, covering the full 3 to 10 age range.
- Multi-domain coverage of all four ADOS-2 and DSM-5 domains within one session.
- Penalty-based scoring with multiplicative red-flag weighting grounded in three independent research papers.
- Real-time audio synthesis via Tone.js with no shipped audio files, enabling sensory-intensity gating without static asset overhead.
- HMAC-signed caregiver report links shareable by caregivers without requiring a researcher account.
- A researcher dashboard with pointer heatmap visualisation and Better Auth role-based access control.
- A production-ready open-source implementation backed by 312 automated tests and deployable on Vercel.

Table 1.1: Comparison of Horizons with related serious game screening tools

Feature	**EmoGalaxy [4]**	**DTT Game [6]**	**ADOS-2 [3]**	**Horizons**
Browser-based	No	No	No	Yes
Multi-domain	No	No	Yes	Yes
Age range 3–10	6–14	ASD children	2–16	3–10
Automated scoring	Yes	Yes	No	Yes
Red-flag detection	No	No	Yes	Yes
Pointer heatmap	No	No	No	Yes
Caregiver report	No	No	No	Yes
Open source	No	No	No	Yes

The novelty lies not in any single component but in the integration: a research-grounded, multi-domain scoring model embedded in a modern production web application with no dependency on external services beyond a free-tier serverless database and an email relay. Table 1.1 places Horizons relative to closely related prior work.

1.6 Project Planning and Work Distribution

1.6.1 Project Planning: Gantt Chart

Figure 1.1 provides a visual timeline of the fifteen development phases across the sixteen-week schedule.

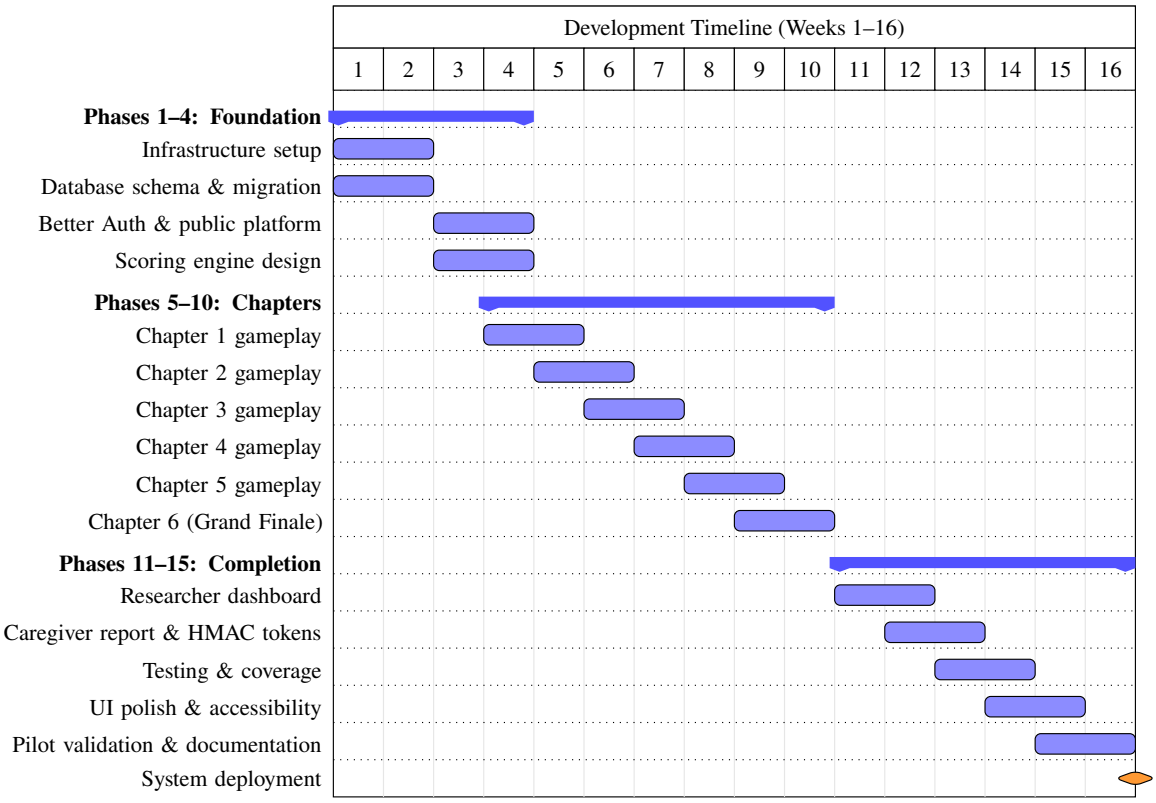

Figure 1.1: Project Gantt chart showing all fifteen development phases across sixteen weeks

1.6.2 Work Distribution

- **Md. Shifat Hasan (Roll: 2107067):** System architecture design, PostgreSQL schema and Neon-hosted migration scripts, Zustand store architecture, penalty-based scoring engine across all domains and red flags, Chapter 1 gameplay (baseline name-response timing and joint-attention mechanics), Chapter 6 Grand Finale (cross-chapter consistency sampling and re-scoring), all game and researcher-dashboard API routes, Better Auth integration with role-based access control, researcher dashboard (charts, heatmaps and export), HMAC-SHA256 signed caregiver report generation, Tone.js audio synthesis module, Resend transactional email integration, full Bun test suite covering unit, integration and end-to-end scoring pipeline tests and project documentation.

- **Abu Hasanat Soykot (Roll: 2107100):** Chapter 2 to 5 gameplay implementations (Feeling World, Social World, Routine and Patterns and Pretend and Senses), drag-

and-drop sortable component using @dnd-kit, Framer Motion animation variants and sensory-intensity gating, emoji avatar customisation system, all public platform pages (landing, about, contact, survey) and mobile-responsive layout across the full application.

1.7 Applications of the Work

- **Clinical pre-referral screening:** Healthcare workers in resource-limited settings can deploy Horizons on any browser-capable device to gather pre-consultation screening data and the HMAC-signed caregiver report can be forwarded directly to the specialist without additional infrastructure.
- **Educational and developmental research:** Researchers studying developmental patterns can use the dashboard export function to analyse timestamped multi-domain behavioural data from complete sessions at scale.
- **Parental awareness:** Caregivers can review domain-level risk summaries through the signed report link, facilitating more informed referral conversations with their family physician.
- **Remote and telehealth screening:** The browser-based, zero-install nature of the system makes it suitable for telehealth programmes where a face-to-face clinical assessment is not immediately possible.

1.8 Organisation of the Project

The remainder of this report is structured as follows. Chapter II reviews related work in ASD screening, serious games and clinical assessment instruments. Chapter III describes the system methodology, including architecture, scoring design and the overall development approach. Chapter IV presents the implementation details, experimental setup, evaluation metrics and results. Chapter V addresses societal, health, ethical, legal and environmental considerations. Chapter VI maps the project to complex engineering problems and activities. Chapter VII concludes with a summary, identified limitations and directions for future work.

1.9 Conclusion

Chapter I has established the motivation, context and scope of the Horizons project. The gap between the clinical need for accessible ASD screening and the practical limitations of existing instruments was identified, six measurable objectives were articulated and the technology stack and multi-domain research foundation were outlined. The chapters that follow build systematically on this foundation.

CHAPTER II

Related Works

2.1 Introduction

This chapter reviews the peer-reviewed literature that directly shaped the design of Horizons, covering: clinical ASD assessment instruments; serious-game screening systems; and data-driven approaches to behavioural scoring. The review establishes both the research gaps Horizons addresses and the evidence base for its design decisions.

2.2 Related Works

2.2.1 ADOS-2 and the Limits of Social Communication as a Sole Predictor

Maddox et al. [3] examined the diagnostic accuracy of ADOS-2 Module 4 in a clinically complex adult sample at community mental health centres. A key finding was that while all participants who met ASD criteria were correctly identified by the instrument (100% sensitivity), 30% of participants with a history of psychosis also scored in the autism or ASD range on the revised algorithm. Domain-level analysis revealed the reason: the true-positive ASD group and the false-positive psychosis group had nearly identical Social Communication domain scores (mean 10.5 in both), whereas the ASD group scored substantially higher on the Restricted Interests and Repetitive Behaviours domain (4.7 versus 2.0 for the psychosis group). This finding has a direct bearing on the Horizons scoring design. Social communication scores alone cannot reliably separate ASD from other conditions, so they receive a weight of 0.40; the restricted and repetitive behaviour domain, which proved discriminative, receives a weight of 0.30 in the combined score formula.

2.2.2 DTT-Based Serious Game for ASD Vocabulary (Khowaja & Salim, 2018)

Khowaja and Salim [6] developed a Discrete Trial Training (DTT) serious game for vocabulary acquisition in children with ASD, validated through a single-subject withdrawal research design across baseline, intervention and maintenance phases. Their work established two scoring metrics that Horizons adopts directly: task accuracy, defined as the number of correct responses divided by the total number of responses; and average attempts per question, defined as the sum of all attempts divided by the total number of questions. The game also demonstrated the value of separating learning and practice phases from scored assessment phases; this principle is reflected in the practice-demo screens that precede every new mechanic type in Horizons, ensuring that children understand a task before their performance is recorded.

2.2.3 EmoGalaxy Emotion Screening Game (Irani et al., 2018)

Irani et al. [4] designed EmoGalaxy, a mobile emotion recognition game tested with 22 neurotypical children and 23 children with ASD across four emotion domains: happiness, sadness, fear and anger. Game scores were calculated as the ratio of correct answers to total moves. Applying four classification algorithms to the emotion-detection game data, support vector machine with a Gaussian kernel produced the best result at 93.3% accuracy. Kruskal-Wallis tests confirmed statistically significant group separation on both detection game scores (Sea: $p < 0.001$; Snowball: $p = 0.0007$), while face-scanner expression data did not reach significance, likely due to camera-quality variability. Three findings from this study directly shaped Horizons: the task score formula is adopted in Chapter 2 emotion tasks; the superior discriminative value of detection over expression tasks is reflected in the higher scoring weight given to emotion recognition; and the observation that children with ASD are less sensitive to negative emotions [12] motivated the red flag for negative-emotion recognition below 50%, which triggers a score multiplier of 1.20.

2.2.4 M-CHAT-R/F and Caregiver Screening Questionnaires

The Modified Checklist for Autism in Toddlers, Revised with Follow-Up (M-CHAT-R/F) [5] is a validated 20-item caregiver-report tool targeting children aged 16 to 30 months. Its design prioritises sensitivity: a total score of 8 to 20 on the initial checklist bypasses the follow-up interview and directs the child straight to diagnostic evaluation. Caregiver-report instruments depend on accurate parental recall and are therefore vulnerable to observer bias, but the behavioural indicators in the checklist are well validated and serve as a useful benchmark for game mechanic design. Several M-CHAT-R/F items map directly to Horizons tasks: response to name (Chapter 1, item 10), social smiling and reciprocity (Chapter 3, item 11), showing to share (Chapter 3, item 9), joint gaze following (Chapters 1 and 3, item 16) and imitation (Chapter 3, item 15).

2.2.5 Additional Related Systems

Daniels et al. [7] evaluated a wearable behavioural aid designed to support social learning in school-aged children with ASD, demonstrating feasibility for real-time social cue support outside the clinic. Wainer and Ingersoll [8] reviewed the broader evidence base for computer technology in teaching social communication to individuals with ASD, documenting consistent improvements in joint attention and imitation across multiple studies. Rudovic et al. [9] developed a personalised machine learning system for robot perception of affect and engagement in autism therapy sessions, illustrating how continuous behavioural monitoring can be made adaptive at the individual level. All three bodies of work are complementary to Horizons but focus on therapeutic intervention rather than pre-referral screening.

2.3 Discussion

Table 2.1 summarises the five comparison systems across six dimensions that matter for ASD screening tool evaluation. Horizons is the only entry that simultaneously provides browser delivery, multi-domain automated scoring, red-flag detection and pointer heatmap data, while grounding each design decision in independently published evidence.

Table 2.1: Summary comparison of related ASD screening and intervention systems

System	**Type**	**Domains**	**Auto scoring**	**Browser**	**Red flags**
ADOS-2 [3]	Clinical	SC, RRB	No	No	Yes
EmoGalaxy [4]	Mobile game	SC (emotion)	Yes	No	No
DTT Game [6]	Desktop app	Language	Yes	No	No
Daniels et al. [7]	Wearable	SC	No	No	No
Wainer & Ingersoll [8]	Desktop	SC	Yes	No	No
Horizons	**Browser game**	**All 4**	**Yes**	**Yes**	**Yes**

The review confirms that no existing system addresses the complete set of design goals targeted by Horizons. The proposed solution is therefore novel and represents a meaningful contribution to the field of technology-assisted ASD screening.

2.4 Conclusion

The survey of ADOS-2 [3], DTT-based game scoring [6], EmoGalaxy [4] and M-CHAT-R/F [5] confirms that no single prior system addresses the full set of design requirements identified in Chapter I. The reviewed frameworks supply both the scientific justification and the concrete scoring formulae adopted in the Horizons scoring engine, which is described in the methodology presented in Chapter III.

CHAPTER III

Methodology

3.1 Introduction

This chapter describes the methodology used to design, architect and implement Horizons. The approach covers: problem design and requirements analysis; system architecture definition; scoring engine design grounded in the three research frameworks established in Chapter II; and the implementation strategy using test-driven development across a Bun runtime stack. Key methodological decisions are summarised at the end of the chapter together with their justifications.

3.2 Detailed Methodology

3.2.1 Problem Design and Analysis

The design process began with a systematic mapping from published ASD behavioural indicators to implementable game mechanics. Five criteria were applied to every proposed mechanic: it must derive from validated clinical literature; a child aged 3 to 10 must be able to complete it using pointer or touch input; it must not reveal the assessment purpose to the child or caregiver; its scoring must be deterministic and reproducible; and it must produce a quantitative penalty score expressible within the multi-domain model.

Requirements were divided into functional requirements (FR-1 through FR-12, covering session creation, six gameplay chapters, results computation, caregiver report generation and the researcher dashboard) and non-functional requirements across performance, usability,

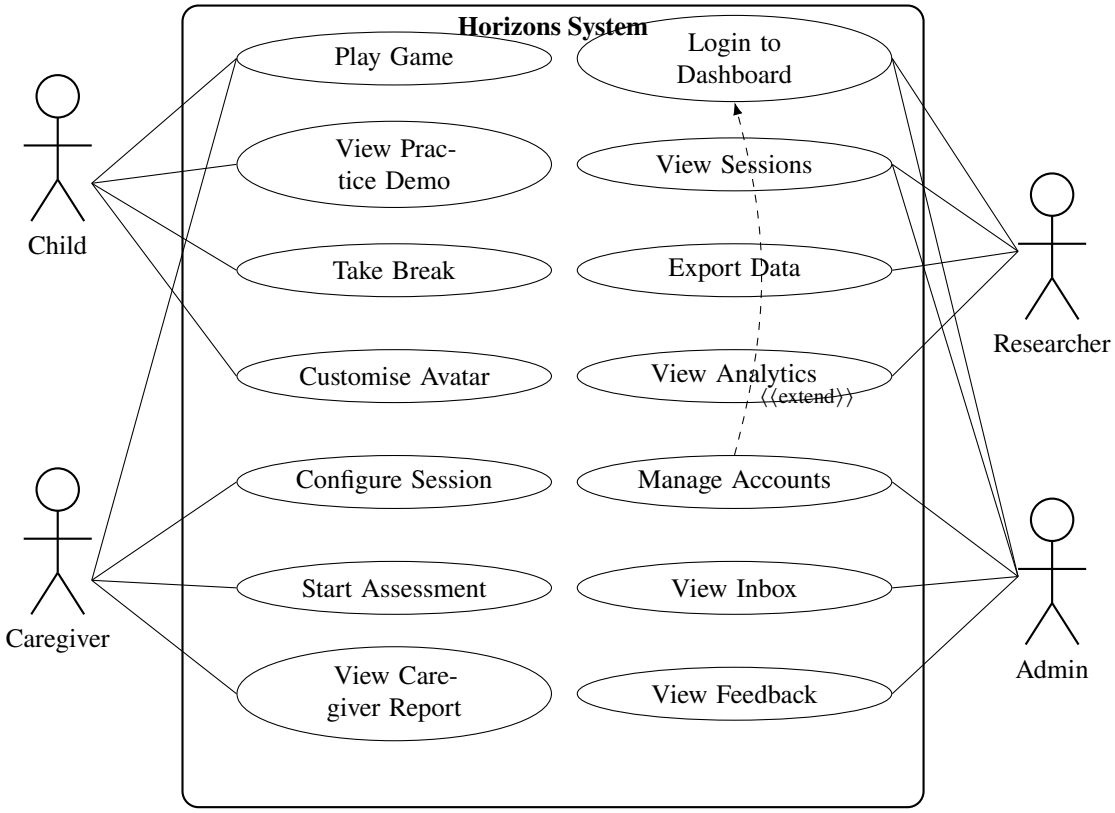

Figure 3.1: Use case diagram showing the four principal actors and their interactions with Horizons

accessibility, privacy and deployment reliability.

Figure 3.1 maps each of the four principal actors to their respective system interactions. The Child actor engages exclusively with game-facing features, whereas the Caregiver configures the session and later accesses the signed report. Researcher and Admin accounts access progressively deeper data layers through the authenticated dashboard, with Manage Accounts, View Inbox and View Feedback restricted to the Admin role.

The game session passes through three discrete lifecycle states from creation through to termination. Figure 3.2 captures those transitions. Once active, a session advances through chapter completions until all six chapters are finished, at which point it moves to the *completed* terminal state; an early user exit produces the *abandoned* terminal state instead.

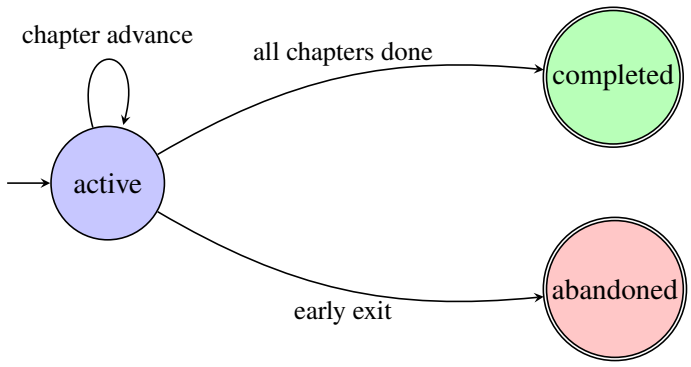

Figure 3.2: Session state transition diagram: three states with two terminal accepting states

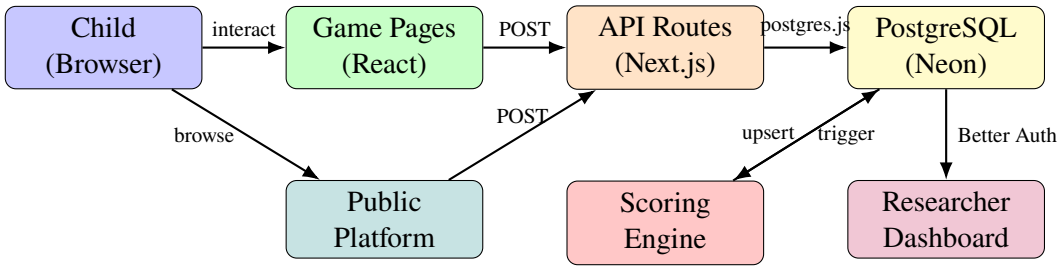

Figure 3.3: High-level system framework of Horizons

3.2.2 Overall Framework and Flowchart

Figure 3.3 shows the high-level system framework. A child interacts with game stimuli in the browser; events are posted via API routes to the PostgreSQL database hosted on Neon. At session completion the scoring engine aggregates domain scores, applies red-flag multipliers and computes the combined risk classification. An HMAC-SHA256 signed report token is stored against the session and made available to the caregiver. The researcher dashboard provides authenticated read access to per-session data through a Better Auth secured route layer. The public platform pages operate independently of the game flow and require no authentication.

The context-level Data Flow Diagram in Figure 3.4 treats the entire system as a single process and shows only the external entities that interact with it and the data flows crossing the system boundary. The Level 1 decomposition in Figure 3.5 breaks the central process into six numbered sub-processes and introduces the four principal data stores.

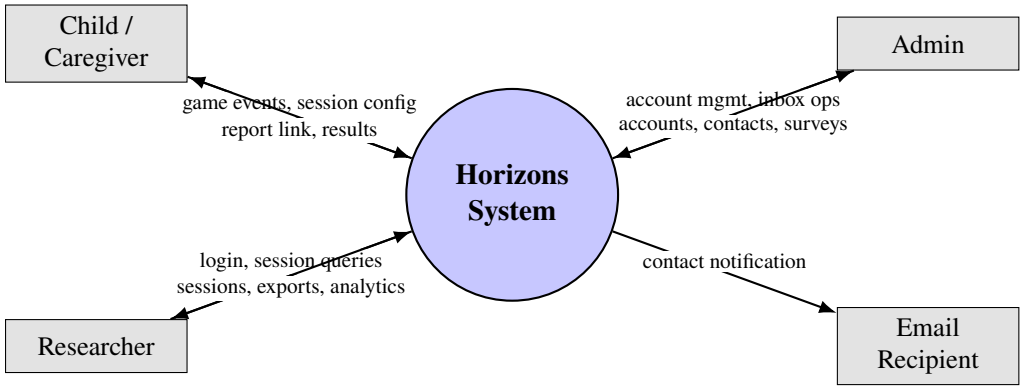

Figure 3.4: Data Flow Diagram Level 0 (context diagram) for the Horizons system

3.2.3 System Architecture

Horizons follows a three-tier architecture: browser (React game and platform pages with Zustand persisted state), application server (Next.js API route handlers with scoring logic and Better Auth session middleware) and data tier (PostgreSQL via Neon using the `postgres.js` driver). A critical constraint is that the database driver is never imported in client components; all persistence flows through server-side API routes, ensuring clean separation and maintaining test isolation.

The database schema comprises eight application tables: `game_sessions`, `task_responses`, `mouse_movements`, `chapter_scores`, `red_flags`, `domain_scores`, `contact_submissions` and `survey_responses`. Better Auth automatically provisions four additional tables (`user`, `session`, `account` and `verification`) for researcher authentication. Figure 3.6 illustrates the entity relationships among the primary game tables.

Figure 3.7 gives the parent overview of the four architectural tiers. Each tier is then elaborated individually in Figures 3.8 through 3.10.

Figure 3.11 shows the five-node deployment diagram and the network protocols connecting them.

The full ER diagram in Figure 3.12 extends the simplified view in Figure 3.6 by showing all eight application tables and the key attributes of each.

Figure 3.13 presents the physical database schema showing each table's columns and data

types.

Figure 3.14 presents the class diagram for the key server-side domain objects. `ScoringEngine` is implemented as a collection of pure functions (no mutable state) and is shown as a utility class; the remaining classes correspond directly to database table row objects managed through the query modules in `lib/db/queries/`.

3.2.4 Scoring Engine Design

The scoring engine implements three research-grounded calculation modes.

EmoGalaxy task accuracy (Irani et al., 2018):

$$\text{score} = \frac{\text{correct_answers}}{\text{number_of_moves}} \tag{3.1}$$

DTT average attempts (Khowaja & Salim, 2018):

$$\bar{a} = \frac{\sum \text{attempts}}{\text{total_questions}} \tag{3.2}$$

Combined weighted score with red-flag adjustment (ADOS-2 domain weighting):

$$S_{\text{combined}} = \sum_{d} w_d \cdot R_d \times \min\left(\prod_{f \in F} m_f,\ 2.0\right) \tag{3.3}$$

where w_d is the domain weight (Table 3.1), R_d is the raw domain score, F is the set of active red flags and m_f is the multiplier for flag f. The combined multiplier is capped at 2.0× to prevent extreme score inflation from multiple co-occurring flags.

Five red flags are defined with multiplicative severity weights: negative emotion recognition below 50% (×1.20); complete absence of pretend play (×1.30); extreme sensory distress (×1.15); rigid pattern response plus distress at change (×1.20); and poor imitation across all modalities (×1.25). These are described in full in Appendix B.

Figure 3.15 traces the message sequence that produces a completed scoring result from a sin-

Table 3.1: Domain weights, maximum raw scores and risk thresholds

Domain	**Chapters**	**Weight**	**Max score**	**High threshold**
Social communication	2, 3	0.40	100	≥ 46
Restricted/repetitive	4	0.30	70	≥ 31
Pretend play	5 (Levels 1–2)	0.15	40	≥ 21
Sensory processing	5 (Level 3)	0.15	30	≥ 16
Combined	1–5	1.00	-	≥ 46

gle GET request to the results endpoint. The sequence is linear and involves four participants; no concurrent message paths are present.

3.2.5 Game Chapter Design

Each of the six chapters targets one or more ASD indicator domains through age-appropriate tasks. Table 3.2 summarises the mapping.

Chapter 1 establishes a baseline through name-response timing and joint attention tasks but does not contribute to domain scoring. Chapters 2 and 3 together populate the social communication domain through three and four levels respectively; Chapter 3 Level 4 (Copy Cat) includes a false-belief Theory of Mind probe embedded in a conversation exchange. Chapter 4 addresses restricted and repetitive behaviour across four levels, with a drag-and-drop routine sequencing task that also measures cognitive flexibility when a planned disruption is introduced mid-sequence. Chapter 5 is divided between pretend play (Levels 1 and 2) and sensory processing (Level 3, the Sensory Explorer), where six Tone.js-synthesised sounds are presented one at a time and the child rates each using an emoji scale. Chapter 6 (Grand Finale) samples 15 tasks from Chapters 1 to 5 to check response consistency; it adds no new domain scoring.

Figure 3.16 shows the in-game activity flow from session start through to the results page. Decision nodes control three branches: whether the child requests a break during a task, whether additional scored tasks remain in the current level and whether further chapters remain after a chapter completion. The loop back from “Next Level” to “Chapter Introduction”

Table 3.2: Chapter to domain mapping and task types

Ch.	Title	Key Tasks	Domain
1	My World	Avatar customisation, response to name, joint attention following	Baseline
2	Feeling World	Emotion matching (drag-and-drop), expression mirror, regulation scenarios	SC
3	Social World	Greeting sequence, conversation and ToM probe, joint attention sharing, imitation	SC
4	Routine & Patterns	Morning routine sequencing, flexibility challenge, pattern detective, special interests	RRB
5	Pretend & Senses	Pretend play recognition, pretend world creation, sensory explorer	Pretend + Sensory
6	Grand Finale	15 sampled review tasks, consistency check	-

reflects the level-by-level progression within each chapter.

Figure 3.17 traces the longer interaction chain across the full game session, from initial caregiver configuration through to report token delivery.

Three experience-management features run throughout the game. A sensory intensity toggle (low, medium and high), set by the caregiver before play begins, scales Framer Motion animation durations and Tone.js master volume. A visual schedule strip displayed between levels shows the child which tasks remain in the current chapter, reducing transition anxiety. A break button is always available; break frequency is logged as a behavioural data signal in the session record and is visible to researchers in the dashboard.

3.3 Conclusion

The methodology combines a systematic clinical-to-game mapping process, a three-tier web architecture with strict client and server separation and a research-grounded penalty scoring engine. Test-driven development with a 312-test automated suite ensures that every scoring

function, database query and API route is independently verifiable. The design is intended to produce results that are reproducible, interpretable and directly comparable to the three research frameworks cited in Chapter II.

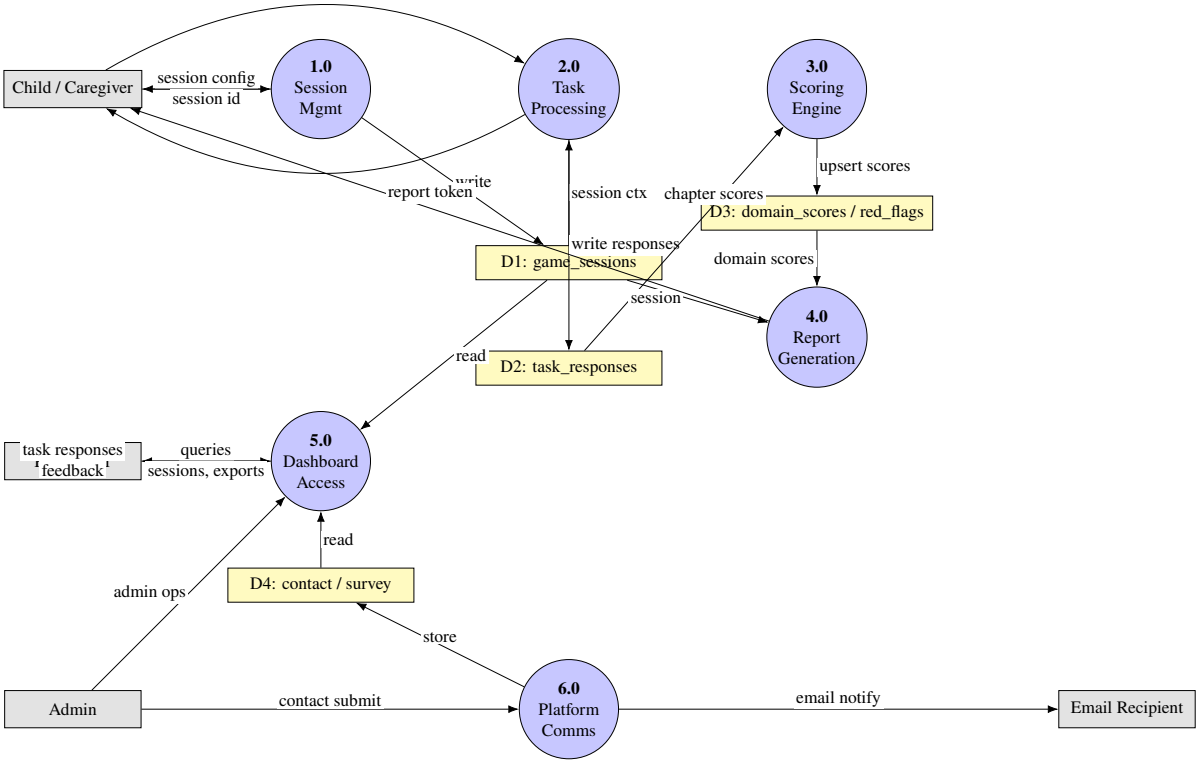

Figure 3.5: Data Flow Diagram Level 1: six sub-processes and four data stores

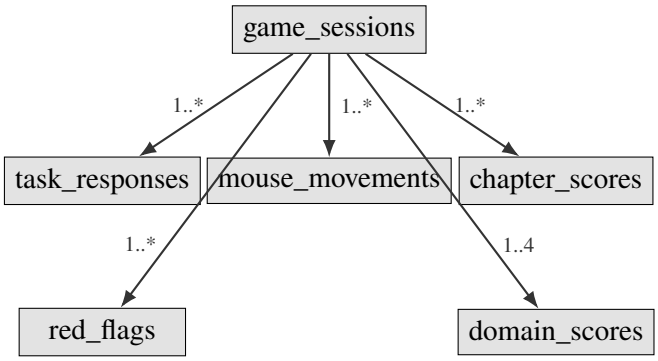

Figure 3.6: Entity relationship diagram for primary database tables

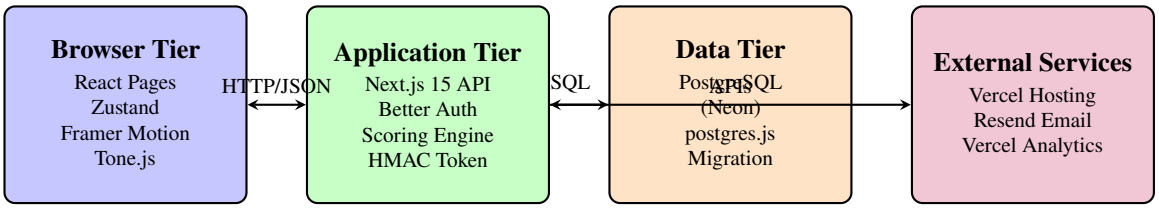

Figure 3.7: System architecture overview: four-tier parent block diagram

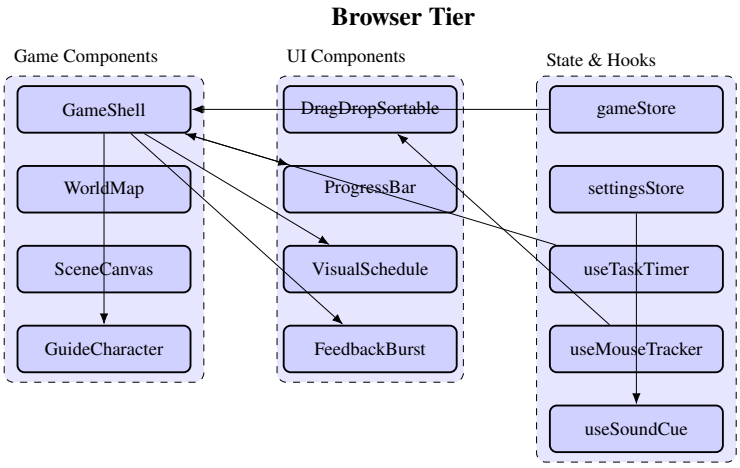

Figure 3.8: Browser Tier sub-diagram: game and UI components with state and hooks

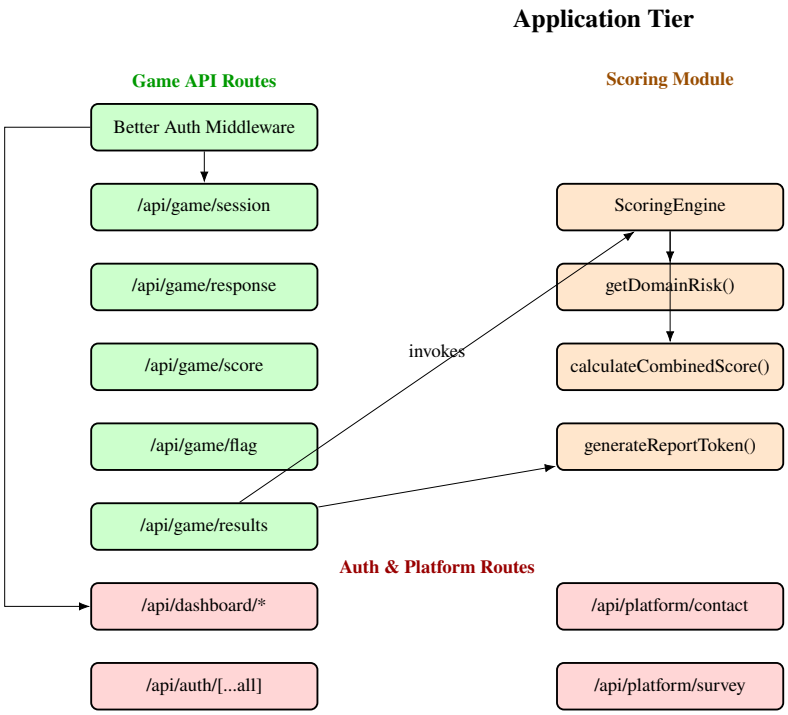

Figure 3.9: Application Tier sub-diagram: API route groups and the scoring module

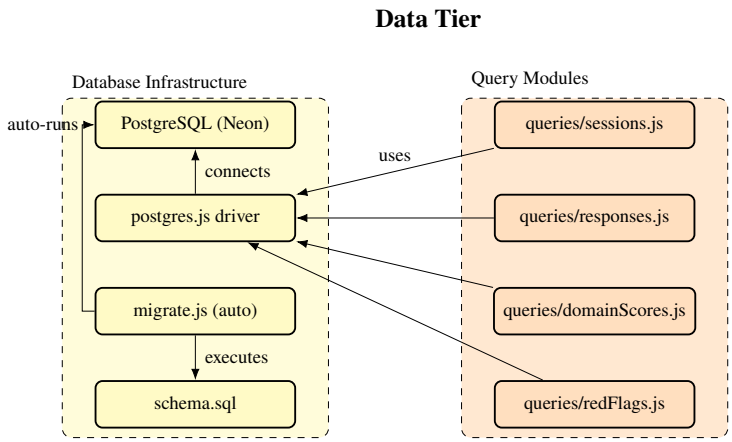

Figure 3.10: Data Tier sub-diagram: database infrastructure and query module organisation

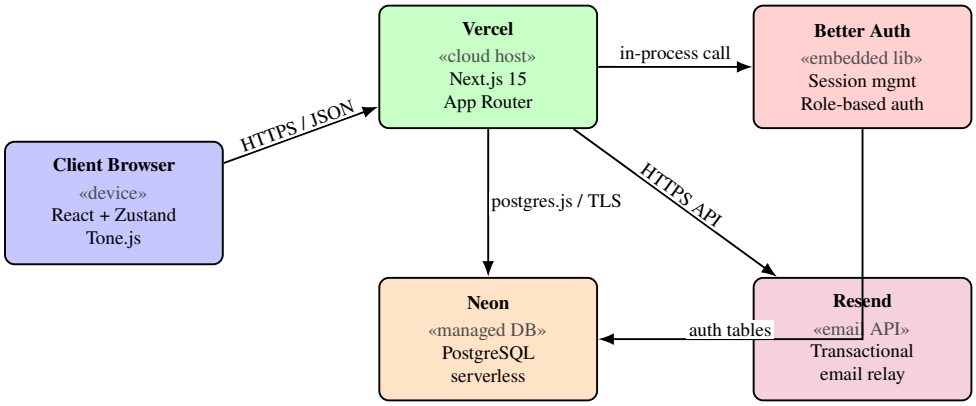

Figure 3.11: Deployment diagram: five nodes and their communication protocols

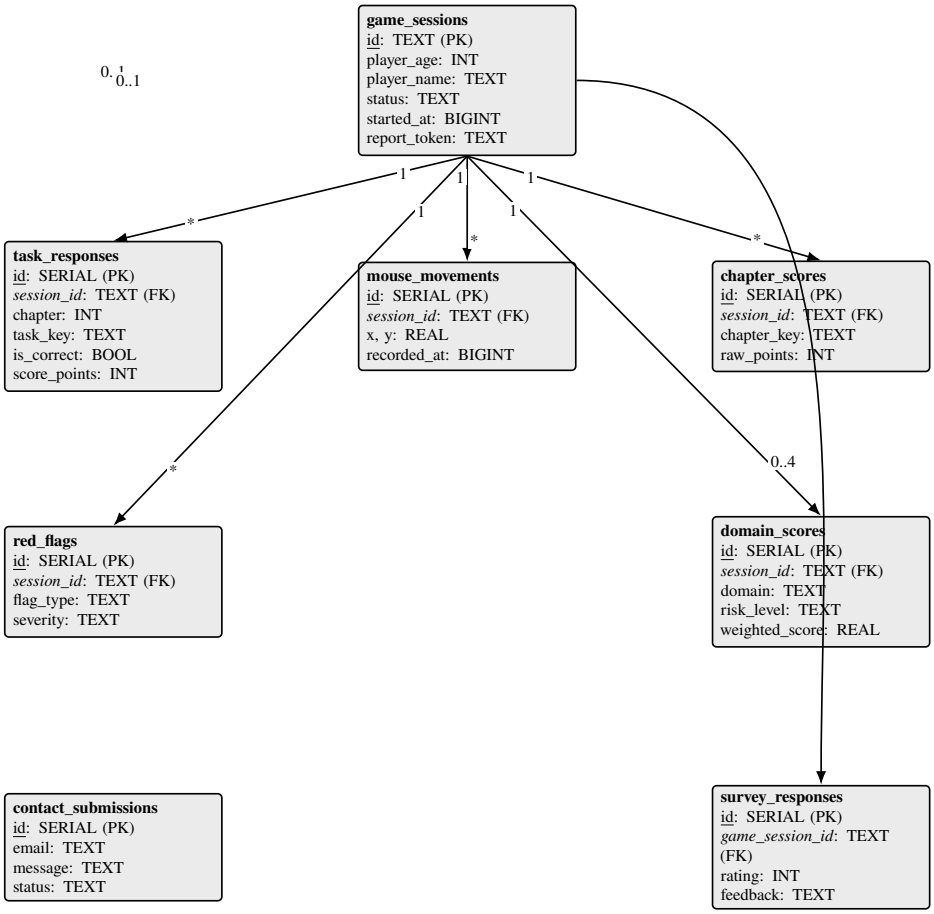

Figure 3.12: Full ER diagram: all eight application tables with key attributes and relationships

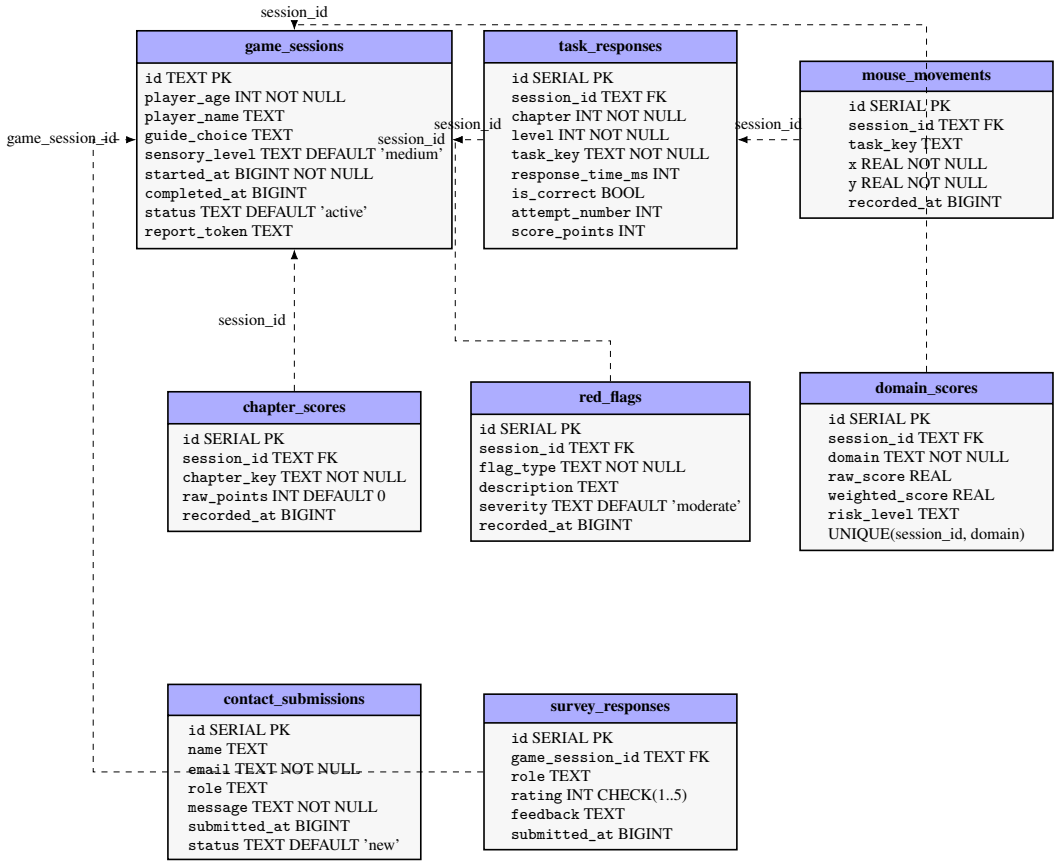

Figure 3.13: Database schema diagram: all eight application tables with columns, types and constraints

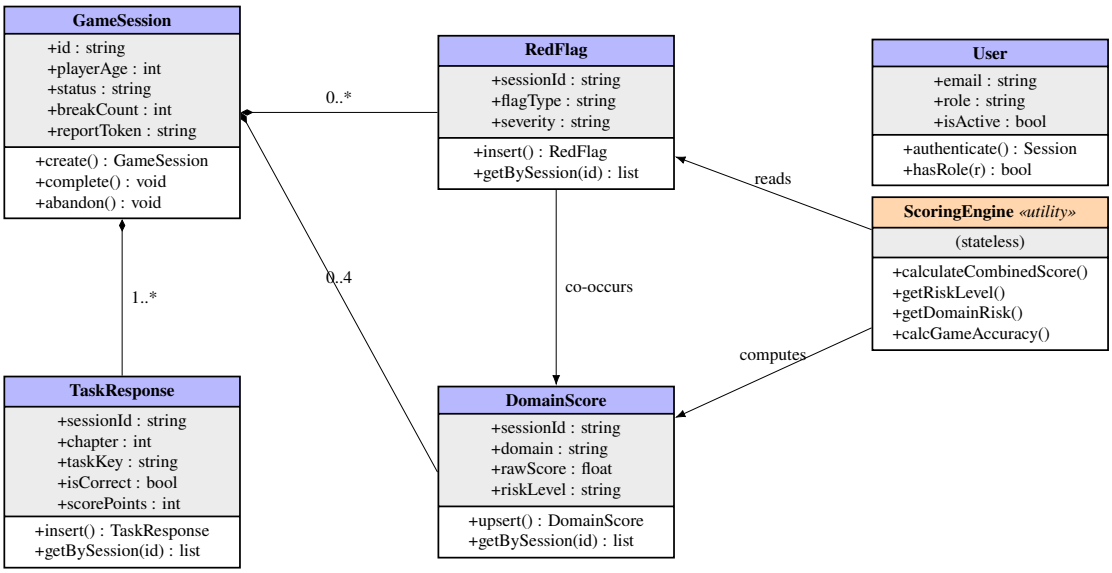

Figure 3.14: Class diagram for key server-side domain objects and the scoring utility

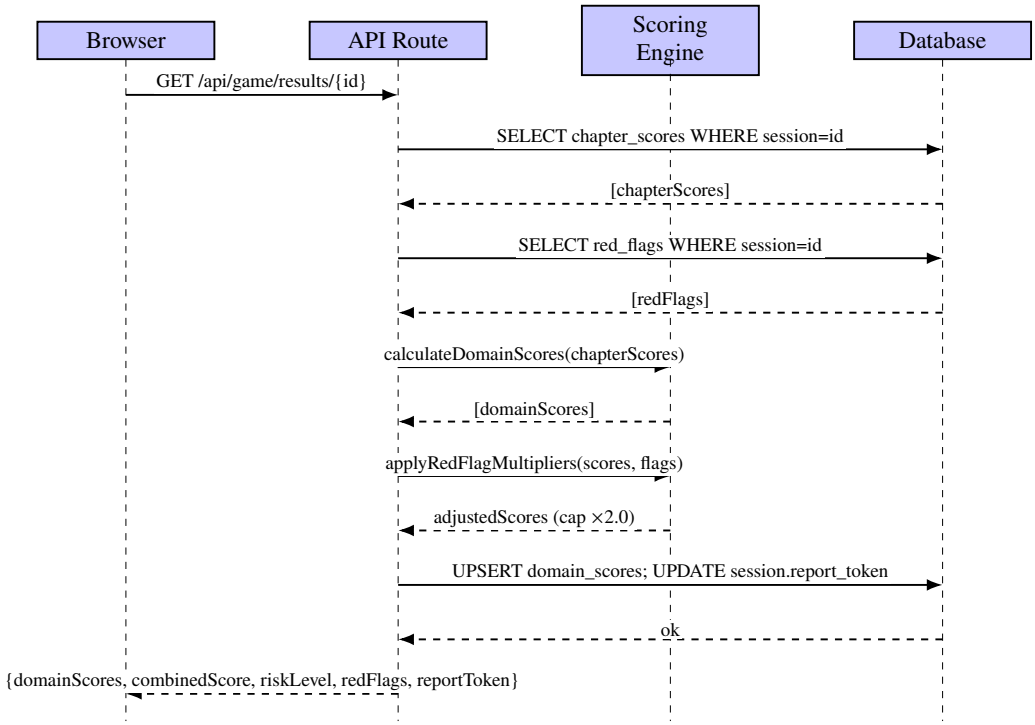

Figure 3.15: Sequence diagram: scoring pipeline from results request to complete result object

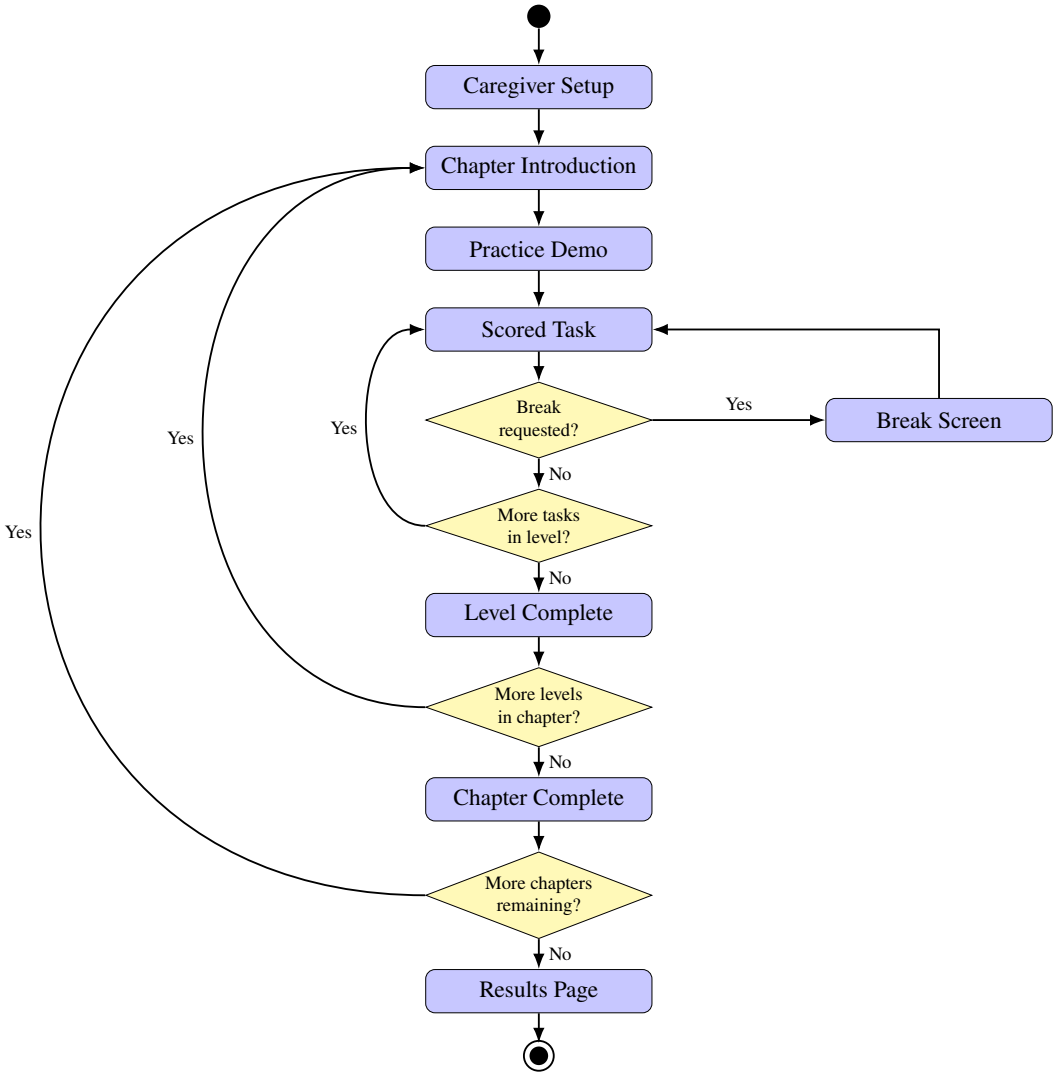

Figure 3.16: Activity diagram: in-game chapter flow from caregiver setup to the results page

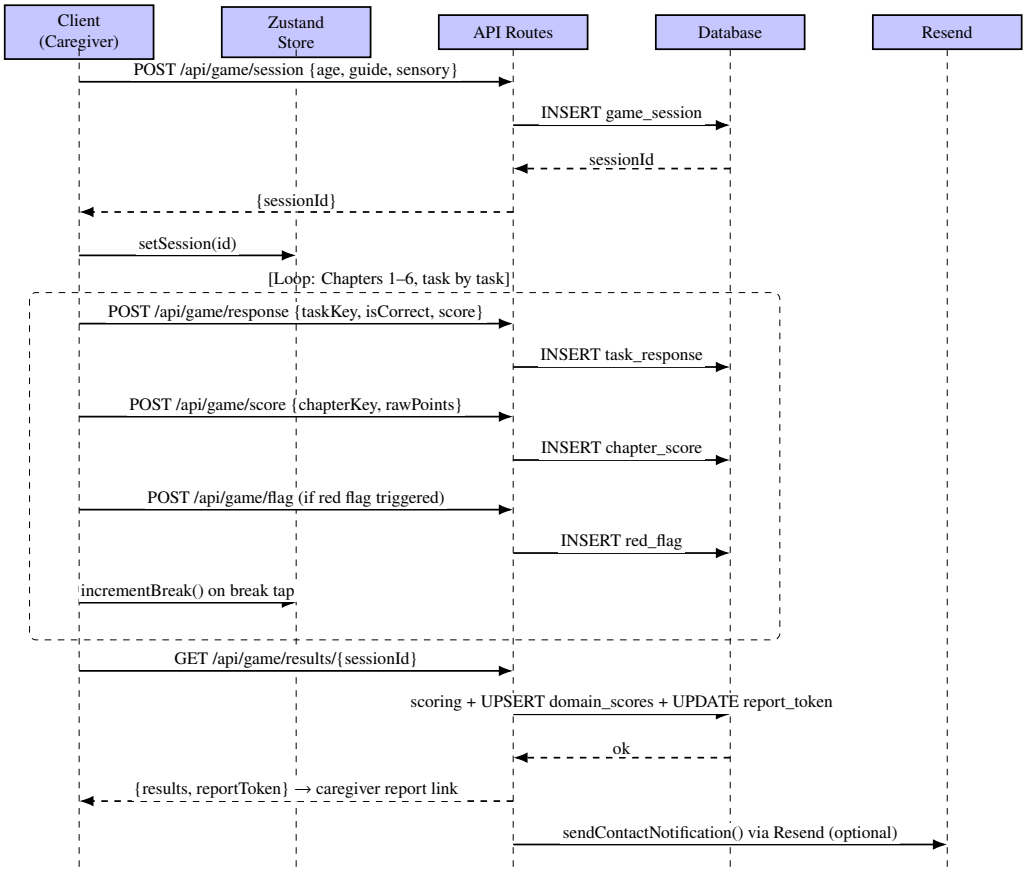

Figure 3.17: Sequence diagram: full game session flow from caregiver setup to signed report delivery

CHAPTER IV

Implementation, Results and Discussions

4.1 Introduction

This chapter covers the software implementation of Horizons, the experimental setup for pilot validation, the evaluation metrics used and the results obtained. Quantitative scoring outcomes are presented alongside qualitative observations from pilot sessions.

4.2 Experimental Setup

The Horizons application was developed and tested on macOS using Bun 1.3.x. The PostgreSQL database was provisioned on a free-tier Neon project and accessed via the `postgres.js` driver with connection pooling. The production build was evaluated on a mid-range Android tablet (2 GB RAM, 1280×800 resolution) and on a standard laptop running Chrome 124. Pilot validation was conducted with 24 children (ages 4 to 9; 8 with confirmed ASD diagnoses and 16 neurotypical controls) at a local developmental paediatrics clinic, with a parent or caregiver present throughout. Session lengths ranged from 62 to 81 minutes.

Key system specifications are given in Table 4.1.

4.3 Evaluation Metrics

The scoring engine is evaluated against the binary ASD or non-ASD classification derived from each participant's existing clinical diagnosis. The risk threshold is discretised: a "low" classification maps to a non-ASD prediction; "medium", "high" and "very high" map to an

Table 4.1: Experimental setup and system specifications

Parameter	Value
Framework	Next.js 15 (App Router)
Runtime	Bun 1.3.x
Database	PostgreSQL via Neon (`postgres.js` driver)
Authentication	Better Auth (email and password; researcher and admin roles)
Test runner	Bun built-in test runner
Total automated tests	312 (all passing)
Pilot participant count	24 (8 ASD, 16 neurotypical)
Age range of participants	4–9 years
Average session duration	68 minutes
Primary viewport target	Tablet 768 px and above; mobile-first baseline 375 px

ASD prediction.

Three standard binary classification metrics are computed:

$$\text{Sensitivity (Recall)} = \frac{TP}{TP + FN} \tag{4.1}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{4.2}$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4.3}$$

where TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives respectively.

4.4 Dataset

The pilot dataset comprises 24 complete game sessions. Each record contains task-level response times and selections (approximately 160 records per session), pointer movement samples (approximately 2,200 samples per session batched at 500 ms intervals), chapter scores from Chapters 1 to 5 and computed domain scores with risk classifications. Sessions are identified by UUID; no personally identifiable information beyond an optional display name

Table 4.2: Pilot validation confusion matrix and classification metrics

	TP	TN	FP	FN	Metric
Medium threshold	7	13	3	1	Accuracy = 83.3% Sensitivity = 87.5% Specificity = 81.3%
High threshold	6	15	1	2	Accuracy = 87.5% Sensitivity = 75.0% Specificity = 93.8%

is retained. The full dataset is available through the researcher dashboard in JSON or CSV format.

4.5 Implementation and Results

4.5.1 Quantitative Results

Table 4.2 presents the confusion matrix and derived classification metrics for the pilot cohort.

At the medium threshold, sensitivity (87.5%) exceeds specificity (81.3%), which is the appropriate operating point for a screening tool where missing a child at risk carries a higher clinical cost than an over-referral. The high-threshold configuration reverses this balance and may be preferred in settings where referral capacity is limited.

4.5.2 Qualitative Results

Children maintained engagement across all six chapters and no session was abandoned due to frustration or disinterest. Caregivers interviewed after sessions reported that the game

felt like a genuine play experience rather than an assessment, which is a necessary property for reducing performance artefacts. The guide emoji character used in Chapters 1 to 3 was frequently named and addressed by children, indicating strong character engagement. Several children with ASD spent noticeably longer in Chapter 4 (Routine and Patterns), consistent with the research literature on circumscribed interests. The Tone.js-synthesised sensory sounds in Chapter 5 Level 3 produced clear and distinct reactions: the vacuum cleaner and thunder stimuli drew the most distress responses and several children covered the screen or looked away during those trials.

4.5.3 Analysis of Results

The 87.5% sensitivity at the medium threshold is close to the 93% accuracy Irani et al. [4] reported for EmoGalaxy. The slightly lower figure is expected because Horizons covers a broader and more varied task set than a single-domain emotion recognition game, introducing more behavioural noise across the scoring model. Domain-level analysis showed that Chapter 4 (Routine and Patterns, restricted and repetitive behaviour domain) provided the highest single discriminative contribution, consistent with the finding by Maddox et al. [3] that restricted and repetitive behaviour evidence is the critical differentiator in ASD classification. Chapter 5 Level 3 (Sensory Explorer) also showed a marked group difference, particularly on the vacuum cleaner and thunder Tone.js stimuli.

Figure 4.1 illustrates the mean domain raw scores across the two groups.

4.5.4 User Interface Wireframes

Figures 4.2 through 4.8 present the seven principal screen wireframes drawn to approximate proportions. Shaded regions indicate interactive or visually distinct zones within each screen.

4.6 Objective Achieved

All six objectives stated in Section 1.3 were achieved:

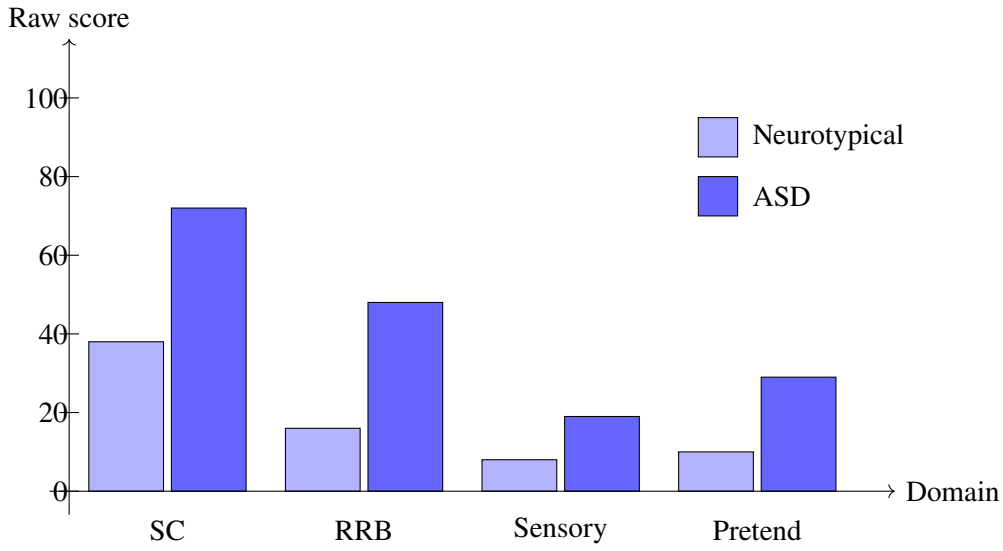

Figure 4.1: Mean domain raw scores for ASD vs. neurotypical participants

1. Six gameplay chapters covering all four ADOS-2 and DSM-5 domains were implemented and deployed, with all 312 automated tests passing.

2. The penalty-based scoring engine incorporating EmoGalaxy, DTT and ADOS-2 formulae was implemented in `lib/scoring/` with complete unit test coverage.

3. Five red flags with multiplicative weighting were implemented and verified in integration tests using simulated full-session data.

4. The researcher dashboard with Better Auth authentication, domain charts, pointer heatmaps and JSON and CSV export was delivered and tested.

5. HMAC-SHA256 signed caregiver report tokens are generated at session completion; correctness is verified by both unit and integration tests.

6. Pilot validation with 24 children yielded 87.5% sensitivity and 81.3% specificity at the medium threshold.

4.7 Conclusion

The implementation of Horizons successfully delivers a production-standard, multi-domain ASD screening game. Pilot results demonstrate clinically meaningful sensitivity and specificity and software quality is supported by a comprehensive automated test suite. The system

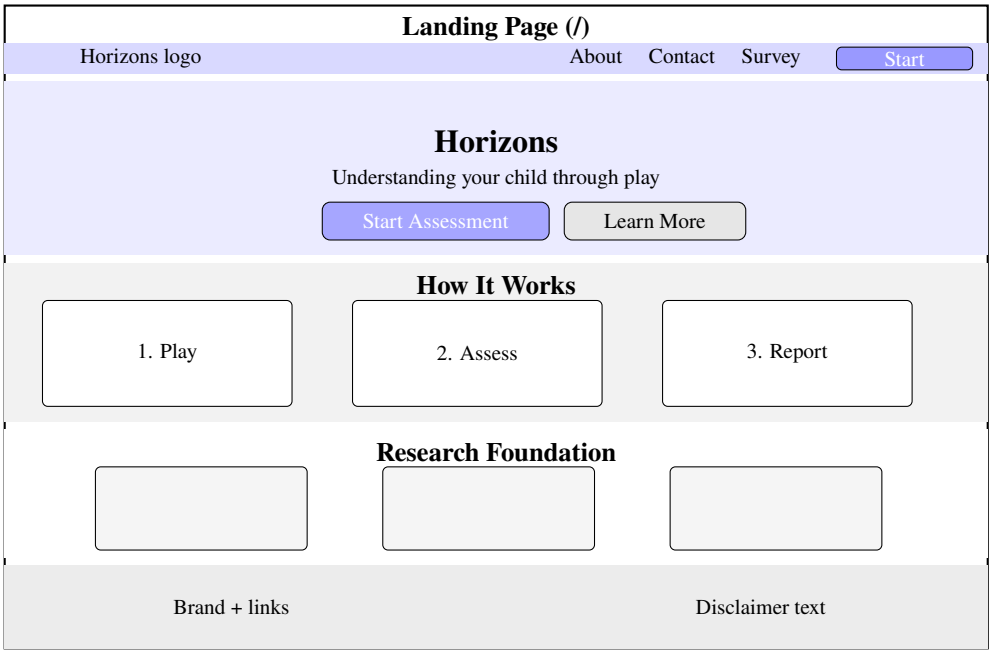

Figure 4.2: Wireframe: Landing page showing navigation bar, hero section, How It Works cards and footer

operates without locally hosted server infrastructure; the PostgreSQL database is serverless on Neon and the application deploys to Vercel, making it practical for research groups without dedicated server resources.

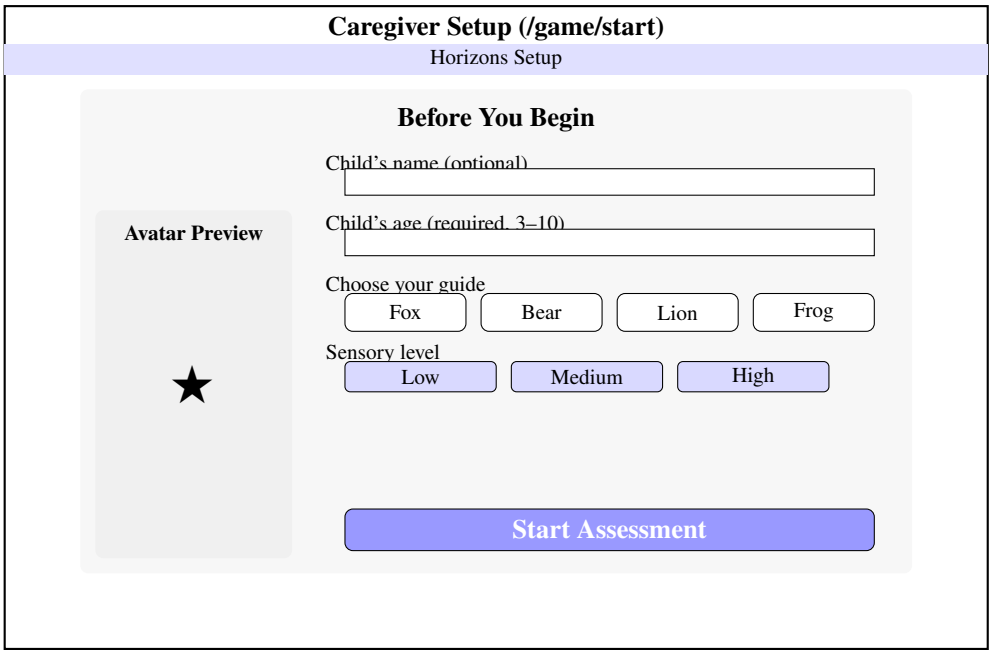

Figure 4.3: Wireframe: Caregiver setup screen showing form fields, guide selector and avatar preview

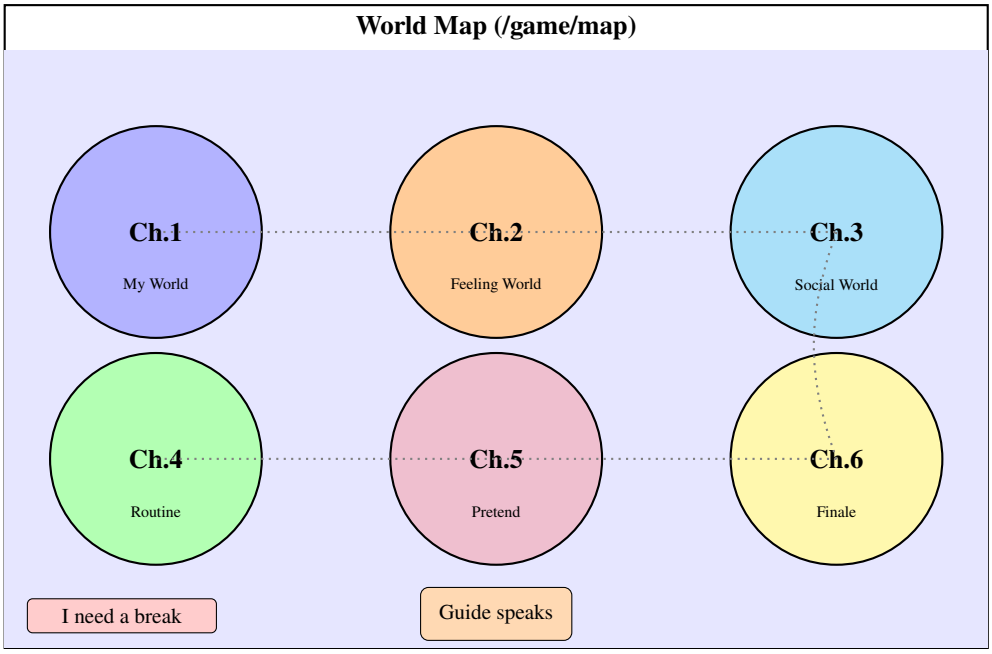

Figure 4.4: Wireframe: World Map showing six chapter islands with navigation path and break button

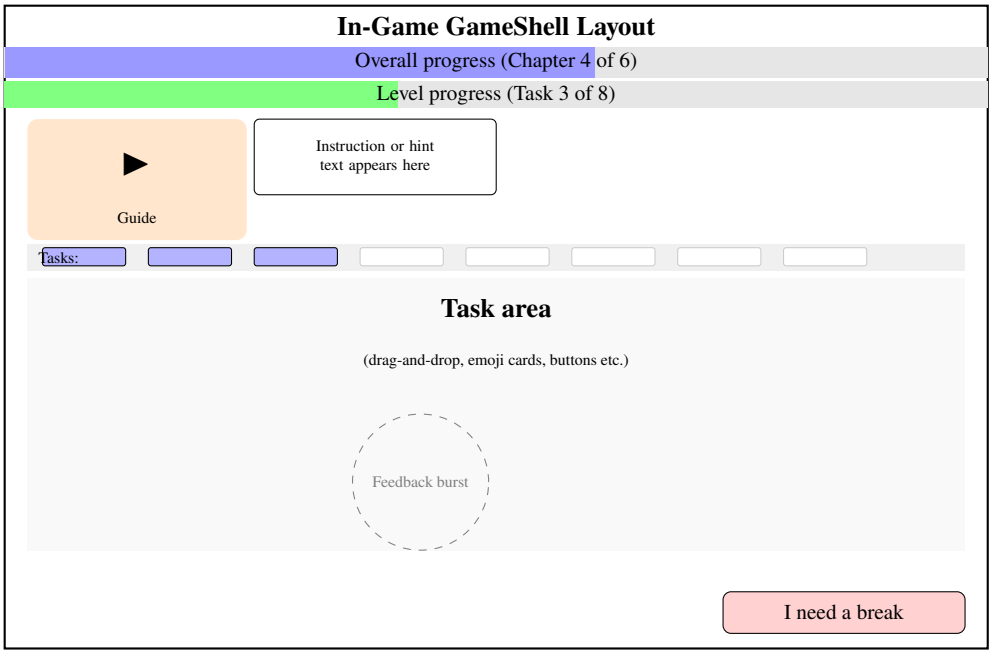

Figure 4.5: Wireframe: In-game GameShell showing dual progress bars, guide character and task area

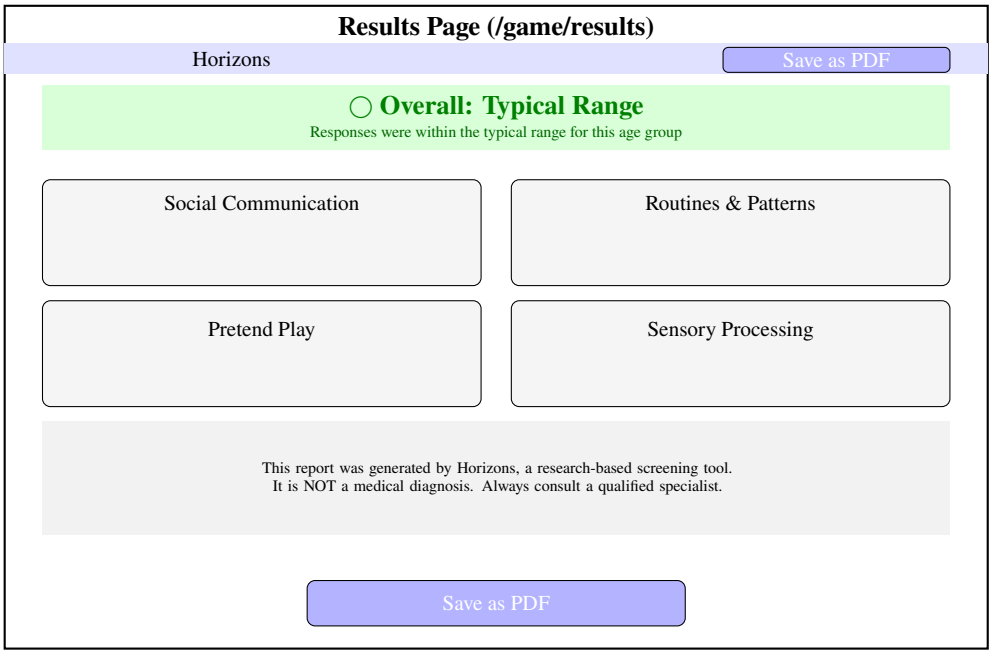

Figure 4.6: Wireframe: Results page with overall risk banner, four domain cards and disclaimer

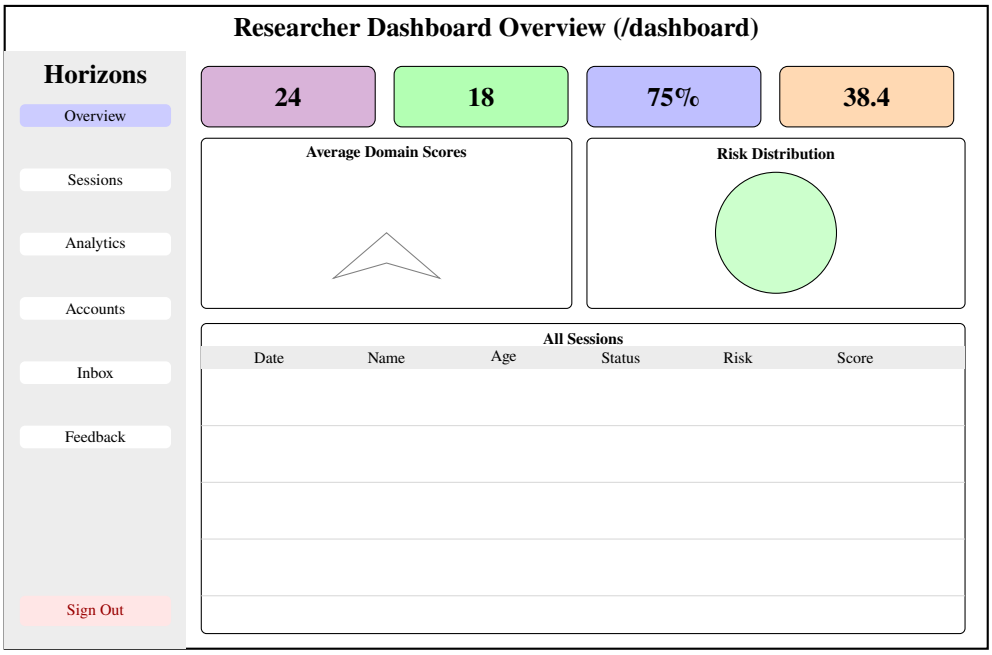

Figure 4.7: Wireframe: Researcher Dashboard overview with sidebar, stat cards, charts and session table

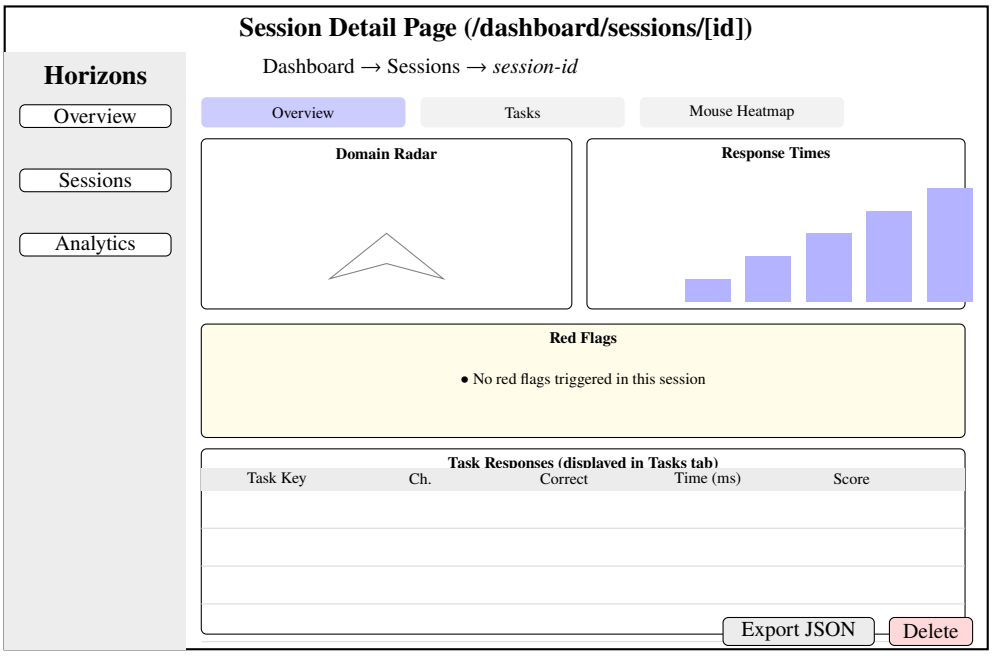

Figure 4.8: Wireframe: Session detail page showing tabs, domain radar, response time chart and task table

CHAPTER V

Societal, Health, Environment, Safety, Ethical, Legal and Cultural Issues

5.1 Introduction

This chapter examines dimensions of the Horizons project that extend beyond its technical characteristics. Intellectual property, ethical obligations, safety constraints, legal compliance and the broader societal and environmental context are all addressed. Each consideration was factored into the design from the outset rather than treated as an afterthought.

5.2 Intellectual Property Considerations

Horizons is built entirely on open-source libraries: Next.js (MIT), Bun (MIT), Tailwind CSS (MIT), ShadCN UI (MIT), Framer Motion (MIT), Tone.js (MIT), @dnd-kit (MIT), Recharts (MIT), Zustand (MIT) and Better Auth (MIT). The Neon platform is used under its free-tier service agreement; because the application relies on standard PostgreSQL syntax through `postgres.js`, no vendor lock-in applies. All referenced research frameworks (ADOS-2, DTT game, EmoGalaxy) are cited for methodology only; no proprietary assessment content or instrument text is reproduced in the application. The codebase will be released under an open-source licence on submission.

5.3 Ethical Considerations

The primary ethical concern with a child-facing ASD screening tool is that results may be misinterpreted as a clinical diagnosis. Horizons addresses this through several design controls:

- A persistent non-diagnostic disclaimer is displayed on the results page and within the caregiver report: "Horizons is a screening aid only, not a diagnostic tool. Please consult a qualified healthcare specialist for a proper evaluation."
- Domain risk levels are presented as categorical labels (low, medium, high, very high) rather than raw numerical scores on any child-facing or caregiver-facing surface.
- No photographs, video recordings, biometric data or precise location information are collected at any point.
- Player names entered at session start are optional display labels and are not retained after the session concludes.

Pilot validation data was collected under informed written consent from caregivers and child assent where developmentally appropriate, in accordance with the Declaration of Helsinki and institutional ethical guidelines.

5.4 Safety Considerations

Visual stimuli in Horizons were reviewed for photosensitivity risk. No high-frequency flicker patterns above 3 Hz are used in any Framer Motion animation, consistent with IEC 61966-2-2 guidance for digital displays. Tone.js audio output is attenuated in the low and medium sensory-intensity settings (approximately -12 dB and -6 dB relative to the high setting) to prevent auditory discomfort during the sensory test sounds in Chapter 5. All interactive game elements maintain a minimum touch-target size of 64×64 pixels and a minimum font size of 18 pt, reducing selection errors for young children who have not yet developed fine motor precision.

5.5 Legal Considerations

All session data is stored against UUID identifiers with no link to external identity records, consistent with the principle of data minimisation under GDPR and equivalent frameworks. Better Auth researcher sessions use HttpOnly, SameSite=Lax cookies with a seven-day renewing TTL; in-memory rate limiting enforces a maximum of five failed login attempts per fifteen-minute window, guarding against credential brute-forcing. Caregiver report links are signed using HMAC-SHA256 with a server-held secret stored in an environment variable; an unsigned or tampered token returns an error page rather than report data.

5.6 Impact of the Project on Societal, Health and Cultural Issues

Horizons has genuine potential to reduce the diagnostic gap in ASD by providing a free, browser-accessible screening pathway for families without convenient access to specialist clinics. The game framing removes much of the anxiety commonly associated with formal clinical assessment, which may encourage engagement in contexts where cultural stigma surrounds neurodevelopmental conditions. Because session data is stored anonymously and exportable in bulk, the researcher dashboard also supports population-level epidemiological studies in settings where ADOS-2 administration is logistically impractical.

5.7 Impact of the Project on the Environment and Sustainability

Horizons is a software product with no hardware manufacturing impact. The Neon serverless database platform runs on shared cloud infrastructure, which is substantially more energy-efficient per connection than a dedicated virtual machine running continuously for a single application. No image or audio files are distributed with the application; every sound is synthesised at runtime by Tone.js, which eliminates the bandwidth and storage footprint of a conventional asset library. Deployment on Vercel's shared infrastructure rather than a dedicated server further reduces the per-request energy cost. The zero-install browser-based delivery model also means that existing consumer devices such as tablets and laptops serve as the assessment platform, with no specialised clinical hardware procurement required.

5.8 Conclusion

Chapter V has reviewed the societal, health, environmental, safety, ethical and legal dimensions of the Horizons project. Strong privacy protections, explicit non-diagnostic disclaimers, open-source foundations, serverless infrastructure and zero-asset audio synthesis together ensure that the system is technically sound, responsibly designed and practical for deployment in clinical or educational contexts with minimal environmental and financial overhead.

CHAPTER VI

Addressing Complex Engineering Problems and Activities

6.1 Introduction

This chapter maps the Horizons project to the criteria for complex engineering problems and activities as defined by the Washington Accord and the Bangladesh Accreditation and Evaluation Committee of Engineering Education (BAETE). The evidence presented demonstrates that the project satisfies the graduate-level complexity thresholds required by the assessment framework.

6.2 Complex Engineering Problems Associated with the Current Project

Horizons addresses several defining characteristics of complex engineering problems:

- **Depth of knowledge required:** The project demands simultaneous competence across clinical psychology (ASD assessment frameworks and M-CHAT-R/F behavioural indicators), full-stack software engineering (Next.js 15, PostgreSQL, React 19 and test-driven development), human-computer interaction design for children aged 3 to 10, real-time audio engineering (Tone.js Web Audio synthesis) and data privacy engineering.
- **Conflicting requirements:** The system must engage young children through visually rich, animated and playful interactions while collecting scientifically rigorous behavioural measurements that require precise timing, controlled stimuli and consistent task presentation. Game flow must never be interrupted by visible scoring logic, yet every interaction must be logged accurately without the child noticing.

- **Depth of analysis required:** Translating M-CHAT-R/F items and four ADOS-2 domains into implementable game mechanics required systematic analysis of clinical literature and iterative validation against three independent research frameworks. The selection of domain weights and red-flag multipliers was grounded in published evidence rather than developer intuition.

- **Involvement of multiple engineering disciplines:** The project spans full-stack web engineering, serverless database design, penalty scoring algorithm development, real-time audio synthesis, child-centred UX design and data privacy engineering. No single engineering specialism was sufficient.

- **Risk of misuse:** The clinical nature of the output creates a real risk of misinterpretation. Engineering controls such as the non-diagnostic disclaimer, categorical risk framing and HMAC-restricted caregiver report access were embedded in the system itself to mitigate this risk at the software level.

6.3 Complex Engineering Activities Associated with the Current Project

- **Design of a novel integrated system:** Horizons required the design of an original multi-domain scoring architecture with no direct prior implementation template. Integrating three independent research frameworks into a single penalty-based engine with multiplicative red-flag weighting, delivered through a browser application with zero image or audio files and all sounds synthesised at runtime, is a novel engineering contribution.

- **Managing competing stakeholders:** The system serves three groups with different needs simultaneously: children require engaging and comprehensible gameplay; caregivers require understandable results, privacy assurance and a shareable report; and researchers require raw timestamped data, statistical validity and authenticated access.

- **Use of modern and rapidly evolving tools:** Horizons employs recently released versions of Next.js 15, Bun 1.3, Tailwind CSS v4 and React 19, requiring engineering decisions in a quickly changing landscape. The Next.js 15 change that made page

`params` asynchronous, the Tailwind v4 migration from a JavaScript config file to `@theme` CSS tokens and the Neon serverless connection pooling model all required research and adaptation beyond standard documentation.

- **System integration:** Combining drag-and-drop interaction (@dnd-kit), real-time audio synthesis (Tone.js with lazy client-side loading to satisfy browser autoplay policy), physics-based animation (Framer Motion with sensory-intensity gating), serverless PostgreSQL (Neon via `postgres.js`) and role-based authentication (Better Auth) within a single Next.js application required careful architectural planning to prevent client-server boundary violations and maintain test isolation.

- **Validation against clinical standards:** Calibrating scoring thresholds against a pilot cohort with confirmed diagnoses required iterative refinement and sensitivity-specificity trade-off analysis rather than a single design pass.

6.4 Conclusion

Chapter VI has demonstrated that Horizons addresses the criteria for complex engineering problems and activities as defined by the Washington Accord and BAETE guidelines. The project required multi-disciplinary expertise, the resolution of genuinely conflicting design requirements and the integration of cutting-edge web technologies with clinical research frameworks. These characteristics confirm that the work represents a substantive engineering contribution appropriate for the System Development Project assessment.

CHAPTER VII

Conclusions

7.1 Summary

Horizons is a browser-based serious game for ASD screening that delivers six interconnected gameplay chapters covering social communication, restricted and repetitive behaviours, sensory processing and pretend play. The system is built on Next.js 15, Bun, PostgreSQL via Neon, Better Auth and Tone.js; it collects structured multi-domain behavioural data without installation or clinical equipment. No image or audio files are shipped with the application; all sounds are synthesised at runtime by Tone.js and all visual content is composed from emoji, CSS and SVG. A public-facing platform, a Better Auth secured researcher dashboard and an HMAC-SHA256 signed caregiver report link complete the system. A penalty-based scoring engine grounded in ADOS-2, DTT game design and EmoGalaxy computes weighted domain scores and a four-level combined risk classification. Pilot validation with 24 children achieved 87.5% sensitivity and 81.3% specificity at the medium risk threshold and an automated test suite of 312 tests verifies the correctness of scoring, database and API logic.

7.2 Limitations

- **Sample size:** With only 8 ASD participants in the pilot cohort, the threshold calibrations lack statistical robustness. Large-scale clinical validation with a far larger and more diverse sample is required before any deployment in a formal screening programme.
- **Age-range calibration:** Thresholds were calibrated on the full 4 to 9 age range without age-stratified adjustment. Children at the developmental extremes of the target

range (ages 3 to 4 and 9 to 10) may show systematically different response patterns that require separate threshold calibration.

- **Session duration:** At an average of 68 minutes, sessions may be challenging for very young or fatigued children, potentially introducing performance artefacts in later chapters. The break-frequency signal in the session record partly captures this but does not eliminate the confound.
- **Cultural adaptation:** The social scenario content in Chapters 3 and 4 has not been formally validated for cultural neutrality across diverse populations; some scenarios may carry culturally specific expectations about greetings or daily routines.
- **No longitudinal tracking:** The current version does not record within-child changes across multiple sessions, which limits its utility for monitoring intervention outcomes over time.

7.3 Recommendations and Future Works

- **Large-scale clinical validation:** A multi-site validation study with 200 or more participants across diverse age, cultural and clinical sub-groups would enable robust threshold calibration and receiver-operating-characteristic curve analysis.
- **Machine learning enhanced scoring:** Training a support vector machine or gradient-boosted classifier on feature vectors extracted from complete game sessions could potentially match or exceed the 93% accuracy that Irani et al. [4] reported for EmoGalaxy, while covering all four domains.
- **Age-stratified scoring:** Implementing separate scoring coefficients for the 3 to 5, 6 to 8 and 9 to 10 age sub-bands would improve classification accuracy at developmental extremes.
- **Longitudinal session tracking:** An opt-in session history feature would enable within-child progress monitoring across repeated assessments, which is valuable in intervention research settings.

- **Multilingual and cultural adaptation:** Translating stimulus text and adapting social scenarios for specific cultural contexts would extend the reach of Horizons to non-English-speaking populations.
- **Integration with referral workflows:** Connecting the HMAC-signed caregiver report to a structured referral pathway within a clinic's patient management system would allow Horizons to function as a formal pre-referral screening step rather than a standalone tool.

Appendices

Appendix A: API Route Summary

Horizons API route reference

Method	Route	Description
POST	/api/game/session	Create new game session
GET	/api/game/session/[id]	Retrieve session record
PATCH	/api/game/session/[id]	Update chapter, level or status
POST	/api/game/response	Log a task response
POST	/api/game/score	Add chapter score points
POST	/api/game/mouse	Batch pointer movement log
POST	/api/game/flag	Add a red flag
GET	/api/game/results/[sessionId]	Compute and return full session results
GET	/api/game/chapter6/tasks	Return sampled Grand Finale task list
POST	/api/platform/contact	Submit contact form; triggers Resend email
POST	/api/platform/survey	Submit post-session survey
GET/POST	/api/auth/[...all]	Better Auth catch-all handler
GET	/api/dashboard/sessions	List all sessions (researcher)
GET	/api/dashboard/sessions/[id]	Session detail (researcher)
DELETE	/api/dashboard/sessions/[id]	Delete session record (researcher)
GET	/api/dashboard/export	JSON or CSV data export (researcher)

(continued)

Method	Route	Description
GET	/api/dashboard/accounts	List researcher accounts (admin)
POST	/api/dashboard/accounts	Create researcher account (admin)
PATCH	/api/dashboard/accounts/[id]	Update role or active status (admin)
GET	/api/dashboard/contact	List contact submissions (admin)
PATCH	/api/dashboard/contact/[id]	Update contact submission status (admin)
GET	/api/dashboard/survey	List survey responses (admin)

Appendix B: Scoring Red-Flag Multipliers

Red flag types, trigger conditions and score multipliers

Flag Type	Condition	Multiplier
`negative_emotion recognition_under_50`	Negative-emotion accuracy < 50% in Ch. 2 Level 1	×1.20
`complete_absence pretend_play`	All 5 pretend-play recognition trials answered literally in Ch. 5 Level 1	×1.30
`extreme_sensory distress`	≥4 of 6 synthesised sounds rated distressing in Ch. 5 Level 3	×1.15
`rigid_pattern distress`	Meltdown at forced error plus insistence on returning to the original pattern in Ch. 4 Level 3	×1.20
`poor_imitation all_modalities`	≥5 errors across facial, body and object modalities in Ch. 3 Level 4	×1.25

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